

Target-Space Seizure Forecasting with a Conditional Cross-Attention Diffusion Model

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Abstract

Forecasting epileptic seizures from brief preictal recordings is difficult because weak pre-onset structure and subject variability limit direct classification. We introduce target-space seizure forecasting, in which a conditional cross-attention diffusion model translates a 3 s preictal window of electroencephalography (EEG), electromyography (EMG), and thalamic calcium-photometry signals into a distribution of onset-adjacent EEG spectrograms. A frozen target-window classifier scores the generated targets, and stochastic samples from independently trained models are averaged to obtain an event probability. We evaluated the approach in reticular thalamic (RT) and thalamocortical (TC) cohorts from a Gnb1 mouse model of epilepsy using fixed mouse-disjoint training, validation, and test partitions across six preictal horizons. In both held-out test subjects, full multimodal diffusion outperformed the matched full multimodal direct classifier at every horizon. EEG-only diffusion remained competitive and sometimes surpassed full multimodal diffusion, particularly in the TC cohort. These findings support target-space generation as a promising subject-disjoint forecasting strategy while showing that the value of multimodal conditioning depends on modality reliability.

Keywords: seizure forecasting; conditional diffusion; electroencephalography; multimodal fusion; cross-attention; domain adaptation

1 Introduction

Epilepsy is characterized by recurrent seizures arising from abnormal brain activity, and approximately one-third of patients remain resistant to antiseizure therapy (Kwan & Brodie, 2000). The unpredictability of these events contributes to injury, psychosocial distress, and loss of autonomy. Reliable seizure forecasting - estimating risk before electrographic onset - could enable timely warnings and preventive interventions (Bou Assi et al., 2017; Kerr et al., 2024). Forecasting is harder than detecting an ongoing seizure, however, because preictal signals are weak and nonstationary, vary across subjects, and can yield

apparently strong results under evaluation protocols that do not adequately control for subject shift (Mormann et al., 2007; Kuhlmann et al., 2018).

Deep learning has improved representation learning from electroencephalography (EEG) (Roy et al., 2019), and convolutional and recurrent models can identify predictive structure in preictal signals (Truong et al., 2018; Tsiouris et al., 2018). Most approaches nevertheless map a preictal window directly to a binary label. This formulation requires a classifier to isolate weak event-related evidence in the same domain in which subject-specific nuisance variation is strongest. In small multimodal animal datasets, the problem is compounded by distribution shift across animals and by the possibility that an additional modality is informative in one cohort but noisy in another.

Diffusion models offer a different formulation. They learn conditional data distributions through iterative denoising (Sohl-Dickstein et al., 2015; Ho et al., 2020) and have been adapted to probabilistic time-series forecasting, imputation, and neurophysiological signal generation (Rasul et al., 2021; Tashiro et al., 2021; Vetter et al., 2024; Torma & Szegletes, 2025). Diffusion has also been applied to multi-channel EEG representation and early seizure warning (Jiang et al., 2026). These studies establish diffusion as a flexible model of complex signals, but they do not address whether a generated onset-adjacent neural representation can serve as an intermediate target for forecasting an event from strictly preictal measurements.

We address this gap with target-space seizure forecasting. Given a 3 s preictal window, a conditional cross-attention diffusion model generates plausible 0-3 s post-onset EEG spectrograms; a frozen target-window classifier scores the generated targets, and the scores are averaged across stochastic samples and independently trained models. This design separates preictal-to-target translation from event readout while using only the preictal condition and random noise at inference.

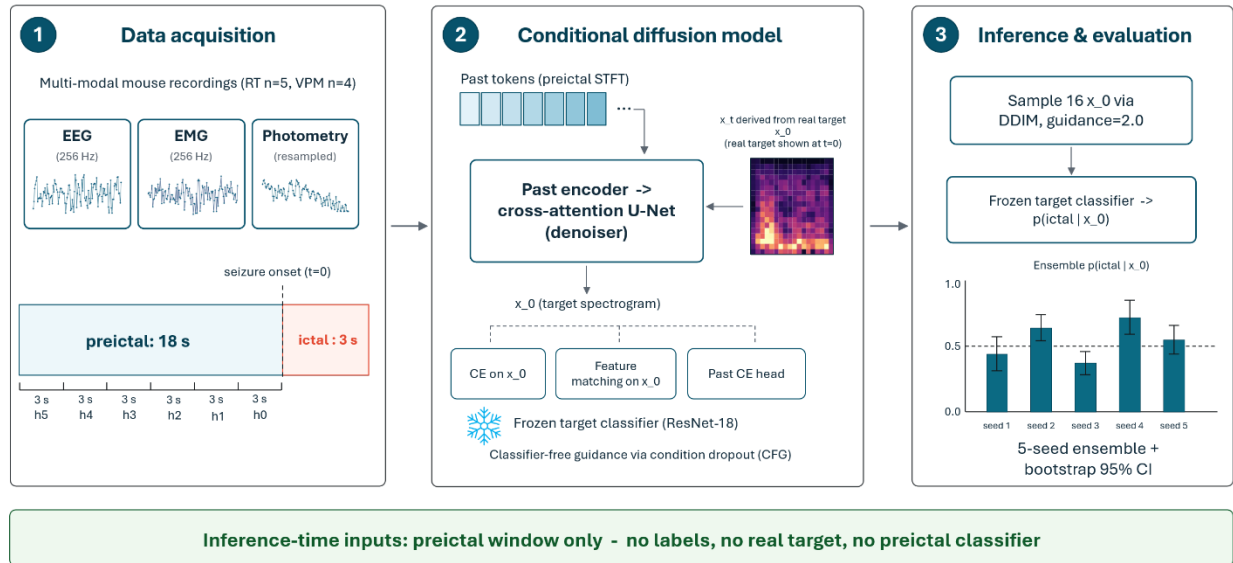


Figure 1. Target-space seizure-forecasting pipeline. Event-centered EEG, EMG, and thalamic calcium-photometry signals are converted into preictal conditions for a cross-attention diffusion model. The model generates multiple post-onset EEG target spectrograms, which are scored by a frozen target classifier and averaged across samples and independently trained models. Inference uses the preictal window and random noise only.

Figure 1 summarizes the pipeline. We evaluate event-centered EEG, electromyography (EMG), and thalamic calcium-photometry recordings from a Gnb1 mouse model of epilepsy. Reticular thalamic (RT) and thalamocortical (TC) recordings are analyzed as separate cohorts under fixed mouse-disjoint training, validation, and test partitions across six non-overlapping preictal horizons spanning -18 to 0 s.

This study makes three contributions. First, it formulates seizure forecasting as conditional generation in an onset-adjacent target space rather than direct preictal classification. Second, it introduces a cross-attention diffusion architecture that conditions target generation on multimodal preictal structure without using labels or real post-onset targets at inference. Third, it provides a mouse-disjoint evaluation that separates the benefit of target-space generation from the benefit - and limitations - of multimodal fusion.

2 Methods

2.1 Data and event construction

The source recordings contain simultaneous EEG, EMG, calcium-photometry signals, and seizure-event annotations from the Gnb1 mouse model characterized by Teng et al. (2022). Seizure onsets were identified from spike-wave discharges in the EEG. We analyze two thalamic recording populations as separate cohorts: inhibitory neurons in the reticular thalamic nucleus (RT) and excitatory thalamocortical neurons (TC). These populations have established roles in absence-seizure synchronization (Futatsugi & Riviello, 1998; McCormick & Contreras, 2001; Makinson et al., 2017; McCafferty et al., 2018). Their separation preserves the distinct recording contexts while allowing the same forecasting protocol to be evaluated in both cohorts.

For each ictal event, the preparation pipeline extracts a 21 s event-centered window comprising 18 s before onset and 3 s after onset. Signals are resampled to 256 Hz by linear interpolation, producing 5,376 samples per modality and placing seizure onset at sample 4,608. The diffusion target is always the 0-3 s post-onset window; the condition is one of six non-overlapping 3 s preictal windows.

An ictal window is retained only when the full 21 s interval lies within a recording and contains no seizure other than the indexed onset. A greedy filter enforces at least 15 s between retained ictal windows. For each mouse, non-ictal windows are sampled from the same recordings to match the ictal count. Candidates are rejected if they overlap a seizure, lie within 60 s of any seizure onset, or occur within 15 s of another accepted non-ictal window. Event rows are shuffled identically across modalities and saved without normalization; all normalization statistics are estimated within the training partition.

Table 1 summarizes the prepared event counts by cohort and mouse.

Table 1. Prepared ictal and matched non-ictal event counts by cohort and mouse.

Cohort	Mouse	Ictal	Non-ictal	Total events
RT	M229	76	76	152
RT	M233	94	94	188
RT	M234	40	40	80
RT	M237	83	83	166
RT	M238	133	133	266
TC	M710	119	119	238

Cohort	Mouse	Ictal	Non-ictal	Total events
TC	M711	73	73	146
TC	M1079	35	35	70
TC	M1080	130	130	260

2.2 Forecasting task

Let i index events, let $y_i \in \{0, 1\}$ denote the event label (1 for ictal and 0 for non-ictal), and let $m \in \{\text{EEG}, \text{EMG}, \text{Ca}\}$ denote the signal modality. If $x_{i,m}(t)$ is modality m for event i at onset-relative time t , the condition for horizon h is

$$c_i^{(h)} = \{x_{i,m}(t) : t \in [a_h, b_h]\}, \quad (1)$$

Here $c_i(h)$ is the multimodal preictal condition and $[a_h, b_h]$ is its onset-relative interval, defined by

$$(a_h, b_h) = (-3(h+1), -3h), \quad h = 0, \dots, 5. \quad (2)$$

The fixed onset-adjacent EEG target is

$$x_{0,i} = x_{i,\text{EEG}}(t), \quad t \in [0, 3). \quad (3)$$

A direct baseline estimates the ictal probability from $c_i(h)$. In contrast, the diffusion model draws S generated targets, denoted $\hat{x}_{0,i}(h, s)$, and the frozen target classifier f_ϕ converts each target into a class probability:

$$\hat{p}_i^{(h)} = \frac{1}{S} \sum_{s=1}^S \text{softmax} \left(f_\phi \left(\hat{x}_{0,i}^{(h,s)} \right) \right)_1. \quad (4)$$

In Equation (4), $p_i(h)$ is the final event score, s indexes generated samples, and the subscript 1 selects the ictal-class softmax probability. No event label, real post-onset target, or direct preictal-classifier score is provided at inference.

2.3 Input and target representations

2.3.1 EEG and EMG spectrograms

Each 3 s EEG or EMG trace $x(t)$ is robustly normalized before time-frequency conversion:

$$x'(t) = \frac{x(t) - \text{median}(x)}{Q_{75}(x) - Q_{25}(x) + \epsilon}. \quad (5)$$

In Equation (5), $\text{median}(x)$ centers the trace, $Q_{25}(x)$ and $Q_{75}(x)$ are its first and third quartiles, and ϵ is a small stabilizing constant. We then compute a Hann-window short-time Fourier transform (STFT) at 256 Hz using a fast Fourier transform (FFT) size of 128 samples and a hop length of 32 samples. If $S(f, \tau)$ is the complex STFT coefficient at frequency f and frame τ , the log-magnitude spectrogram is

$$X(f, \tau) = \log(1 + |S(f, \tau)|). \quad (6)$$

Frequencies above 60 Hz are discarded. A 3 s window therefore produces 31 frequency bins and 27 time frames, so each EEG or EMG subwindow is represented by a 31×27 map.

2.3.2 Calcium-photometry channel

Calcium photometry varies more slowly than EEG and EMG, so the photometry trace is not transformed with the STFT. Instead, its samples are averaged into the same 27 time bins used by the spectrograms:

$$b_t = \frac{1}{|I_t|} \sum_{j \in I_t} z_j. \quad (7)$$

In Equation (7), z_j is photometry sample j , $I(\tau)$ is the set of sample indices in time bin τ , and $b(\tau)$ is the corresponding bin mean. The 27-bin sequence is then standardized within the event:

$$\tilde{b}_t = \frac{b_t - \bar{b}}{\sigma_b + \epsilon}, \quad (8)$$

In Equation (8), the numerator subtracts the event-level mean, the denominator uses the event-level standard deviation, and ϵ prevents division by zero. The standardized sequence is broadcast across 31 frequency rows to obtain a spectrogram-shaped channel that varies over time but is constant across frequency.

2.3.3 Normalization

Diffusion inputs and targets are represented in a shared normalized log-magnitude space. The global training-partition mean and standard deviation are applied as

$$\hat{X} = \frac{X - \mu_{\text{train}}}{\sigma_{\text{train}} + \epsilon}. \quad (9)$$

In Equation (9), the training-partition mean and standard deviation are estimated from the gradient-training mice only; the same values are applied to validation and test mice. For the frozen target classifier and direct baselines, each spectrogram-like channel is instead min-max scaled to $[0, 1]$ within sample and resized to 224×224 by bilinear interpolation. The EEG-only classifier receives one channel, whereas the full multimodal preictal classifier receives EEG, EMG, and photometry channels.

2.4 Conditional target-space diffusion model

2.4.1 Overview

The proposed model has three stages. A past encoder first converts the preictal tensor into spatial conditioning tokens. A conditional diffusion model then denoises a noisy target spectrogram while attending to those tokens. Finally, a frozen target-window EEG classifier maps generated targets to event probabilities.

The EEG-only condition has shape $1 \times 31 \times 27$, and the full multimodal condition has shape $3 \times 31 \times 27$ with EEG, EMG, and photometry channels. The target is always a one-channel 31×27 post-onset EEG spectrogram.

Figure 2 shows the conditioning pathway and cross-attention denoiser used in the final experiments.

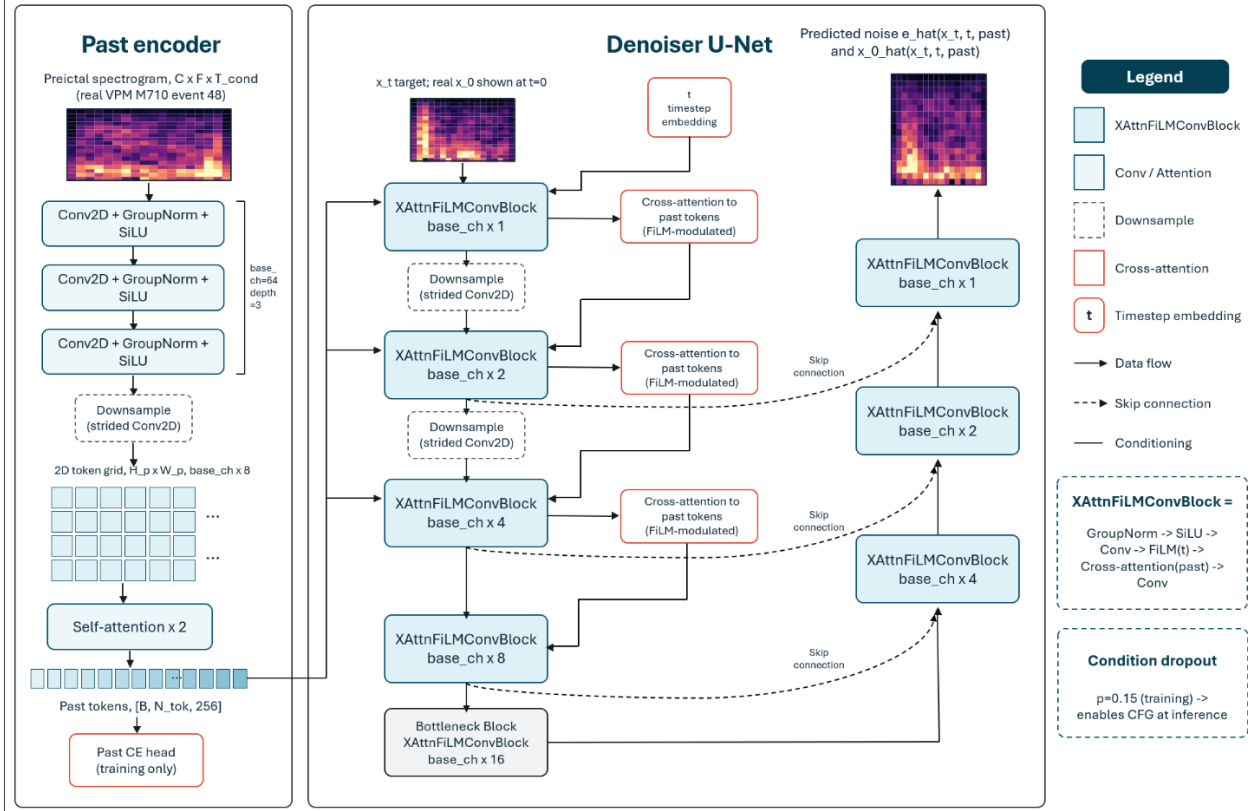


Figure 2. Cross-attention conditional diffusion architecture. The past encoder maps the preictal multimodal tensor to spatial tokens. At each U-Net resolution, noisy target features receive timestep modulation and attend to the same past-token memory. The bottleneck adds target self-attention, and skip connections preserve multiscale target structure. Auxiliary target classification, feature matching, preictal classification, and mouse-adversarial losses are used during training only. The implementation uses Gaussian error linear unit (GELU) activations.

2.4.2 Past-token encoder

The past encoder receives the condition defined in Equation (1), with shape $C \times 31 \times 27$. A 32-channel convolutional stem is followed by three stride-2 blocks that increase the representation to a 256-dimensional token space. A 1×1 projection maps the final feature map to token dimension 256. The projected map is flattened into a spatial sequence and processed by two Transformer-style blocks, each containing multi-head self-attention and a multilayer perceptron (MLP) residual sublayer.

The three stride-2 blocks reduce the 31×27 input to an approximately 4×4 token grid. These tokens preserve coarse time-frequency structure and form the conditioning memory used by every denoising block. A pooled token representation also feeds two training-only heads: a binary past-classification head and a mouse-identity adversarial head.

2.4.3 Cross-attention U-Net denoiser

The denoiser is a two-dimensional U-Net (Ronneberger et al., 2015) operating on the noisy target spectrogram. Its base width is 64, depth is 4, timestep and token dimensions are 256, and the down-path widths are 64, 128, 256, and 512. Each denoising block contains two convolution-group-normalization-

GELU layers, feature-wise linear modulation (FiLM; Perez et al., 2018), a residual connection, and cross-attention from spatial target features to past tokens.

For block activation h , the timestep embedding t produces scale $\gamma(t)$ and shift $\beta(t)$, and the FiLM update is

$$h \leftarrow h \odot (1 + \gamma(t)) + \beta(t). \quad (10)$$

Cross-attention follows the scaled dot-product formulation (Vaswani et al., 2017). Spatial target features h supply queries $Q(h)$, past tokens c supply keys $K(c)$ and values $V(c)$, and d is the key dimension:

$$A(h, c) = \text{softmax}\left(\frac{Q(h)K(c)^T}{\sqrt{d}}\right)V(c). \quad (11)$$

The attention output $A(h, c)$ is added through a learned scalar gate parameter:

$$h \leftarrow h + \tanh(\theta_{\text{gate}})A(h, c). \quad (12)$$

The gate is initialized to 0.5 so that preictal conditioning contributes from the start of training. Multi-head self-attention is also applied at the bottleneck. The denoiser output is the predicted noise $\epsilon\theta(x_t, t, c)$.

2.4.4 Diffusion process

The forward diffusion process corrupts a clean target spectrogram x_0 at timestep t with standard Gaussian noise ϵ :

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (13)$$

Here t is sampled uniformly from $T = 1,000$ diffusion steps, the cumulative coefficient in Equation (13) is the product of the stepwise signal-retention coefficients, and ϵ is drawn from $\mathcal{N}(0, I)$. We use the cosine noise schedule of Nichol and Dhariwal (2021).

The denoiser predicts $\hat{\epsilon} = \epsilon\theta(x_t, t, c)$, from which the clean target estimate $\hat{x}_0$ is reconstructed as

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \hat{\epsilon}}{\sqrt{\bar{\alpha}_t}}. \quad (14)$$

At inference, deterministic denoising diffusion implicit model (DDIM) updates (Song et al., 2020) reduce generation to 50 reverse steps. For the next timestep $t' < t$,

$$x_{t'} = \sqrt{\bar{\alpha}_{t'}} \hat{x}_0 + \sqrt{1 - \bar{\alpha}_{t'}} \hat{\epsilon}. \quad (15)$$

The generated clean-target estimate is clamped to $[-3, 3]$ in normalized space.

2.4.5 Classifier-free guidance

During training, the past condition is replaced by zeros with probability 0.15, which jointly trains conditional and unconditional denoising paths. At sampling, classifier-free guidance (Ho & Salimans, 2021) combines their noise estimates:

$$\hat{\epsilon} = \epsilon_{\theta}(x_t, t, \emptyset) + g[\epsilon_{\theta}(x_t, t, c) - \epsilon_{\theta}(x_t, t, \emptyset)], \quad (16)$$

In Equation (16), the empty-set symbol denotes the zero condition and g is the guidance scale, set to 2.0 in the final experiments. Guidance acts only on the preictal condition and does not use event labels, real targets, or classifier gradients at inference.

2.5 Training objective and regularization

2.5.1 Training objective

The training objective combines frequency-weighted noise prediction, target-classifier cross-entropy, target feature matching, auxiliary past classification, and mouse-adversarial regularization:

$$\mathcal{L} = \lambda_{\epsilon} \mathcal{L}_{\epsilon} + \lambda_{\text{CE}} \mathcal{L}_{\text{targetCE}} + \lambda_{\text{FM}} \mathcal{L}_{\text{FM}} + \lambda_{\text{past}} \mathcal{L}_{\text{pastCE}} + \lambda_{\text{adv}} \mathcal{L}_{\text{mouse}}. \quad (17)$$

The noise-prediction component is a frequency-weighted mean squared error (MSE):

$$\mathcal{L}_{\epsilon} = \mathbb{E}_{t, \epsilon} [w(f) \|\epsilon - \epsilon_{\theta}(x_t, t, c)\|_2^2], \quad (18)$$

In Equation (18), the expectation is over diffusion timestep t and sampled noise ϵ , $\|\cdot\|_2$ is the Euclidean norm, and $w(f)$ decreases linearly from 2.0 at the lowest retained frequency to 0.5 at the highest.

The target-classifier cross-entropy (CE) passes the predicted clean target through the frozen EEG target classifier shown in Equation (19):

$$\mathcal{L}_{\text{targetCE}} = \text{CE}(f_{\phi}(\hat{x}_0), y). \quad (19)$$

This auxiliary loss is applied only for timesteps t in $[100, 900]$, where the clean target estimate is informative without being trivially close to the observed target. The feature-matching loss compares the frozen classifier's encoder representation of the predicted target with that of the real target:

$$\mathcal{L}_{\text{FM}} = \|f_{\phi}^{\text{enc}}(\hat{x}_0) - f_{\phi}^{\text{enc}}(x_0)\|_2^2. \quad (20)$$

The past-classification term trains a pooled-token head to predict event label y . The mouse-adversarial term applies a gradient-reversal layer (GRL; Ganin et al., 2016) before a mouse-identity classifier:

$$\text{GRL}(z) = z, \quad \frac{\partial \text{GRL}}{\partial z} = -\lambda_{\text{GRL}} I. \quad (21)$$

In Equation (21), z is the pooled token representation, λ_{GRL} controls the reversed gradient, and I is the identity matrix. This term encourages the encoder to retain event-relevant preictal information while suppressing mouse identity.

The final loss weights are listed in Table 2.

Table 2. Loss-term weights used in the final diffusion-training configuration.

Term	Weight
Noise-prediction MSE	0.25
Target-classifier cross-entropy	2.0
Target feature matching	10.0
Past-classification cross-entropy	1.0
Mouse-adversarial cross-entropy	0.5

2.5.2 Cross-mouse mixup and time-frequency augmentation

With probability 0.5 per batch, the training loop applies mixup (Zhang et al., 2018). When cross-mouse mixup is enabled, the permutation is retried so that paired samples come from different mice. The interpolation coefficient λ is drawn from Beta(0.4, 0.4), and an input x is mixed as

$$x_{\text{mix}} = \lambda x_a + (1 - \lambda)x_b. \quad (22)$$

The target and preictal tensors use the same interpolation coefficient. The hard event and mouse labels are taken from the dominant sample: sample a when $\lambda \geq 0.5$ and sample b otherwise. The final runs also apply SpecAugment-style masking (Park et al., 2019) to the preictal tensor using two frequency masks of maximum width 6 and two time masks of maximum width 4.

2.6 Target classifier and generated-target scoring

The frozen target classifier is a ResNet18 (He et al., 2016) modified for one-channel 224×224 spectrograms. Its initial convolution is replaced by a one-channel layer initialized from the mean of the ImageNet red-green-blue (RGB) filters when pretrained weights are used. The classifier head contains a 512-to-256 linear layer, a rectified linear unit (ReLU), dropout of 0.3, and a final two-class linear layer. The target classifier is trained on real 0-3 s post-onset EEG spectrograms from non-test mice and is then frozen.

For scoring, a generated target is inverted from global z-score space to log-magnitude space, min-max scaled within sample, resized to 224×224 , and passed through the frozen target classifier. For each event and checkpoint, 16 target samples are generated and their ictal probabilities are averaged. The reported diffusion probability then averages five independently seeded checkpoints, for 80 generated targets per event.

2.7 Training and fixed evaluation splits

Architecture and optimization settings are fixed across the six horizons. The diffusion model uses base width 64, U-Net depth 4, timestep and token dimensions 256, past-encoder depth 3, two past self-attention layers, 1,000 diffusion steps, 50 DDIM steps, guidance scale 2.0, and condition-dropout probability 0.15. Training uses AdamW with learning rate 10^{-4} , weight decay 0.01, cosine learning-rate decay, batch size 128, gradient clipping at 1.0, exponential moving-average (EMA) decay 0.995, and at most 300 epochs. The checkpoint with the highest validation generated-target area under the receiver operating characteristic curve (ROC AUC) is retained. EEG-only and full multimodal diffusion differ only in the number of conditioning channels.

Table 3 gives the fixed subject-level partitions. Validation and test mice are excluded from gradient training, and the test mouse is not used to select checkpoints, estimate normalization statistics, or choose hyperparameters.

Table 3. Fixed subject-level training, validation, and test partitions.

Cohort	Gradient-training mice	Validation mouse	Test mouse	Train events	Validation events	Test events
RT	M233, M234, M237	M229	M238	434	152	266
TC	M711, M1080	M1079	M710	406	70	238

The direct EEG and multimodal classifiers use the same untouched test mouse and the same horizon-specific inputs as the corresponding diffusion models. Classifier checkpoint selection uses the same subject-level validation partition as diffusion training. Comparisons are therefore matched by test subject, event label, horizon, and modality set.

2.8 Evaluation protocol

The primary metric is ROC AUC; accuracy at threshold 0.5 is secondary. Diffusion AUC is computed from ensemble probabilities obtained from five validation-selected checkpoints per horizon and 16 generated targets per checkpoint. Bootstrap 95% confidence intervals (CIs) are computed from saved event-level probability arrays. For the paired effect-size analysis in Figure 3, event-level delta AUC compares each diffusion ensemble with its matched preictal classifier on the same held-out events. CIs use stratified event-bootstrap resampling, and p-values use paired permutation tests with Benjamini-Hochberg false-discovery-rate correction across the planned horizon-level comparisons.

Two modality-matched comparisons are reported: EEG-only diffusion versus the EEG-only preictal classifier, and full multimodal diffusion versus the EEG-EMG-calcium preictal classifier.

The target-window EEG classifier is included only as a reference for the separability of the real post-onset target. It is not a deployable preictal method because it observes the 0-3 s post-onset EEG window.

3 Results

3.1 Main results

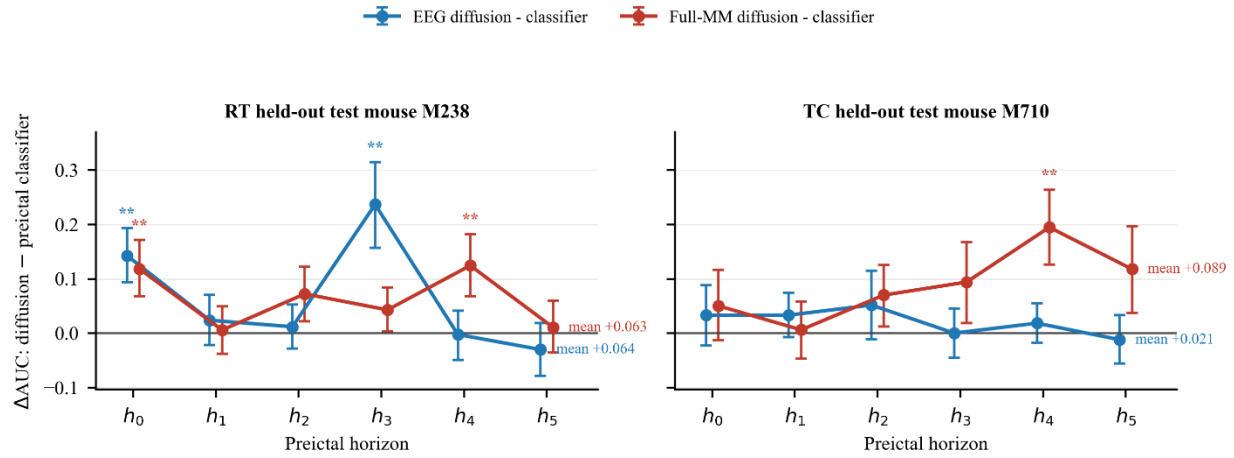
Table 4. ROC AUC on held-out test mice across six 3 s preictal horizons.

Cohort/test mouse	Method	h0	h1	h2	h3	h4	h5	Mean
RT/M238	Target classifier	0.999	0.999	0.999	0.999	0.999	0.999	0.999
RT/M238	EEG classifier	0.763	0.852	0.861	0.595	0.861	0.839	0.795
RT/M238	EEG diffusion	0.906	0.876	0.873	0.832	0.858	0.809	0.859
RT/M238	Full multimodal classifier	0.785	0.856	0.816	0.834	0.768	0.844	0.817
RT/M238	Full multimodal diffusion	0.903	0.862	0.888	0.877	0.892	0.855	0.880
TC/M710	Target classifier	0.923	0.923	0.923	0.923	0.923	0.923	0.923
TC/M710	EEG classifier	0.715	0.731	0.707	0.755	0.749	0.735	0.732
TC/M710	EEG diffusion	0.748	0.764	0.758	0.755	0.768	0.723	0.753
TC/M710	Full multimodal classifier	0.726	0.723	0.684	0.620	0.548	0.576	0.646
TC/M710	Full multimodal diffusion	0.777	0.729	0.754	0.714	0.742	0.694	0.735

Table 4 summarizes the primary AUC results. Figures 4 and 5 show the corresponding ROC curves, and Figure 3 reports paired effect sizes. Full multimodal diffusion outperforms the matched full multimodal direct classifier at every preictal horizon in both cohorts, although the strongest modality setting differs between RT and TC. The RT test subject also has higher target separability and higher mean diffusion AUC than the TC test subject, indicating more consistently decodable seizure-related information in the RT preictal windows.

To assess whether these table-level improvements are stable at the event level, Figure 3 compares diffusion with the matched preictal-window classifier using paired delta-AUC estimates.

Paired event-level AUC gain over matched preictal-window classifiers



Error bars: stratified event-bootstrap 95% CI. Stars: paired permutation test, BH-FDR corrected.

Figure 3. Paired event-level AUC gain over matched preictal-window classifiers. Each point is the AUC difference between a diffusion ensemble and its matched preictal classifier on the same held-out events. Error bars are stratified event-bootstrap 95% CIs; stars indicate paired permutation tests after Benjamini-Hochberg false-discovery-rate correction. RT includes 133 ictal and 133 non-ictal test events; TC includes 119 ictal and 119 non-ictal test events.

In RT, EEG-only diffusion has large corrected gains at h0 and h3, whereas full multimodal diffusion has corrected gains at h0 and h4. In TC, the clearest corrected gain occurs for full multimodal diffusion at h4, while several positive deltas have CIs that include zero. Thus, Table 4 shows a consistent directional advantage of target-space diffusion over matched classifiers, but Figure 3 shows that statistical support depends on horizon and modality.

3.2 RT cohort

The RT evaluation in held-out mouse M238 shows strong target separability and diffusion performance across horizons. The target classifier reaches AUC 0.999 on the real 0-3 s post-onset EEG window. Full multimodal diffusion reaches AUCs of 0.903, 0.862, 0.888, 0.877, 0.892, and 0.855 across h0-h5; the matched full multimodal classifier reaches 0.785, 0.856, 0.816, 0.834, 0.768, and 0.844.

The full multimodal diffusion gain is positive at every RT horizon, with mean gain +0.063 AUC. The largest gain is at h4 (+0.124), and the smallest is at h1 (+0.006). EEG-only diffusion is also strong, with mean AUC 0.859, but it falls slightly below the EEG-only classifier at h4 and h5. The horizon profile is nonmonotonic, supporting sustained horizon-wise forecasting rather than a simple near-onset detection effect.

RT cohort, held-out test mouse M238

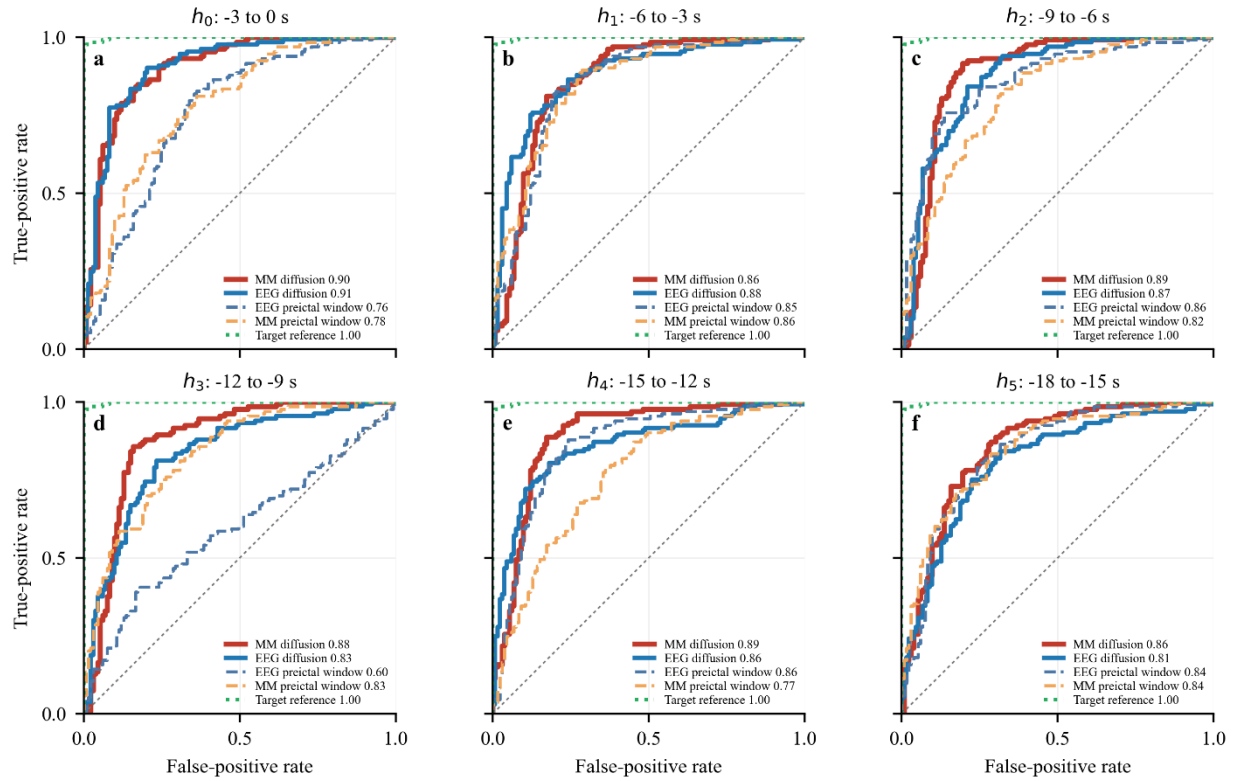


Figure 4. ROC performance for the RT test mouse M238. Each panel corresponds to one 3 s preictal horizon. Solid curves show diffusion models, dashed curves show matched direct preictal classifiers, and the dotted green curve shows the real post-onset target reference. Diffusion probabilities average 16 generations from each of five independently trained models.

3.3 TC cohort

The TC evaluation in held-out mouse M710 is more challenging. The target classifier reaches AUC 0.923, lower than in the RT test subject. Full multimodal diffusion reaches AUCs of 0.777, 0.730, 0.754, 0.714, 0.743, and 0.694 across h_0 - h_5 ; the matched full multimodal classifier reaches 0.726, 0.723, 0.684, 0.620, 0.548, and 0.576.

The full multimodal diffusion gain is positive at all six TC horizons, with mean gain +0.089 AUC. The advantage is largest at h_4 and h_5 , where the full multimodal classifier is weak but diffusion remains above 0.69 AUC. EEG-only diffusion reaches mean AUC 0.753 and exceeds full multimodal diffusion at h_1 - h_5 . This result bounds the paper's central claim: target-space diffusion improves over the matched classifier, but adding EMG and photometry is not uniformly beneficial in the current TC cohort.

TC cohort, held-out test mouse M710

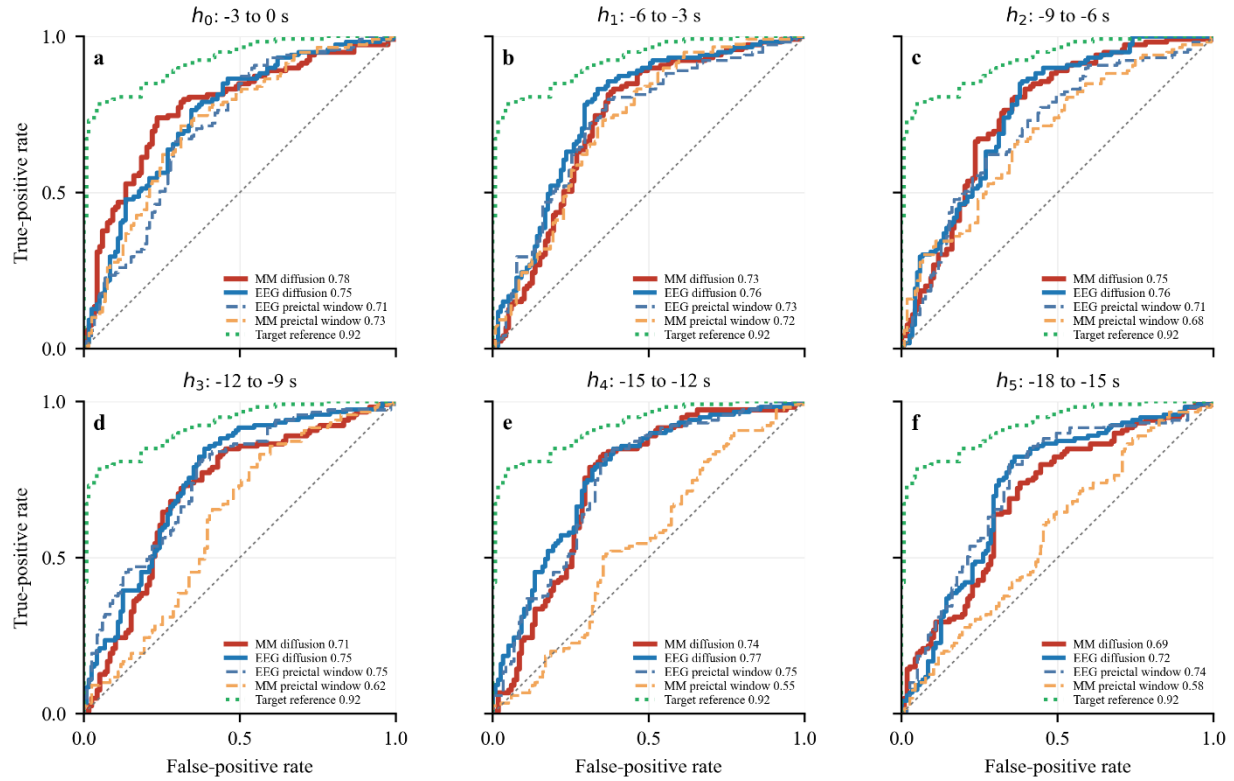


Figure 5. ROC performance for the TC test mouse M710. The display follows Figure 4. Full multimodal diffusion exceeds the matched full multimodal classifier at every horizon, whereas EEG-only diffusion is stronger than full multimodal diffusion at several horizons.

3.4 Generated-target structure and feature-space diagnostics

Figures 6 and 7 summarize class-conditional target structure over the complete held-out test sets. Every real and generated spectrogram is min-max normalized within sample, matching target-classifier preprocessing. For diffusion, one sample is drawn per event from each of five validation-selected checkpoints and averaged within event before class statistics are computed. This preserves the event as the independent unit. The first two columns show class means, the third shows the standardized ictal-minus-non-ictal contrast at each time-frequency bin, and the fourth shows frequency-wise Cohen's d with an event-level bootstrap 95% CI.

RT cohort, held-out test mouse M238: class-conditional h0 target structure

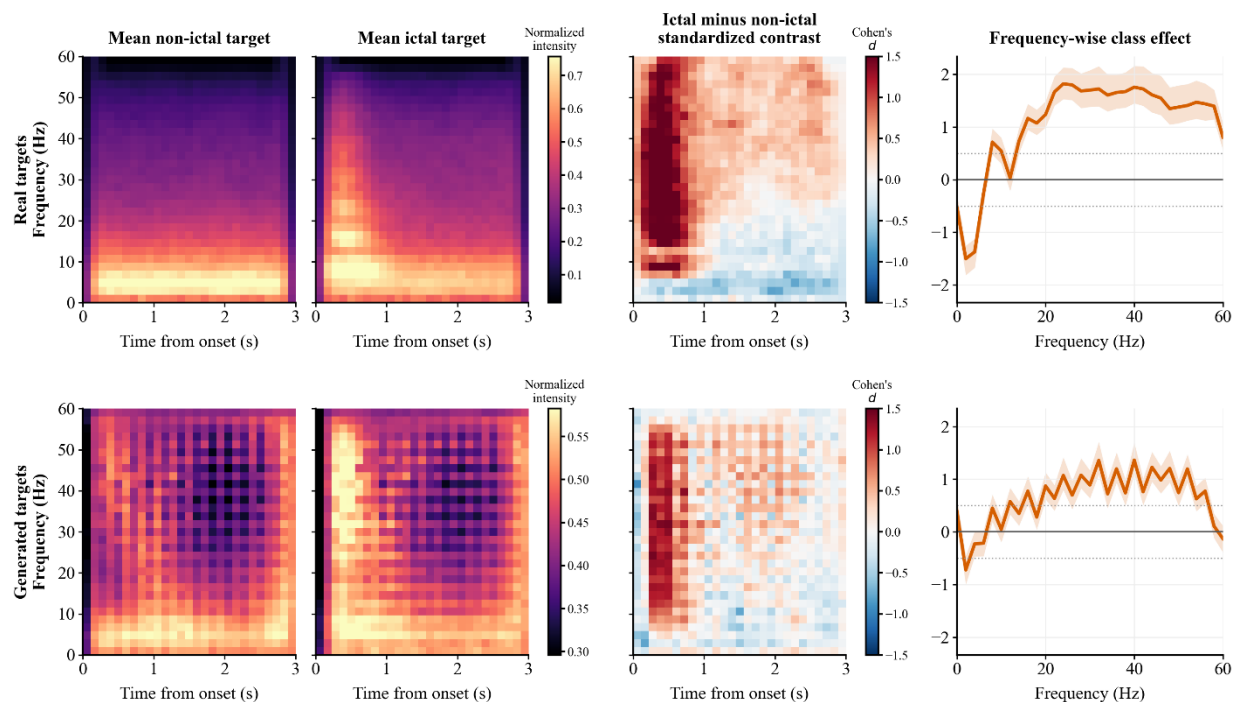


Figure 6. Class-conditional RT target structure at h0. Rows compare real and generated targets for test mouse M238. Non-ictal and ictal means share a color scale within each row. Contrast maps and frequency profiles report ictal-minus-non-ictal Cohen's d ; profile shading is an event-level bootstrap 95% CI. Generated statistics use one sample per event from each of five checkpoints, averaged within event before class aggregation. Generated targets recover a broad positive mid-to-high-frequency ictal contrast, although it is weaker than the real-target contrast.

TC cohort, held-out test mouse M710: class-conditional h0 target structure

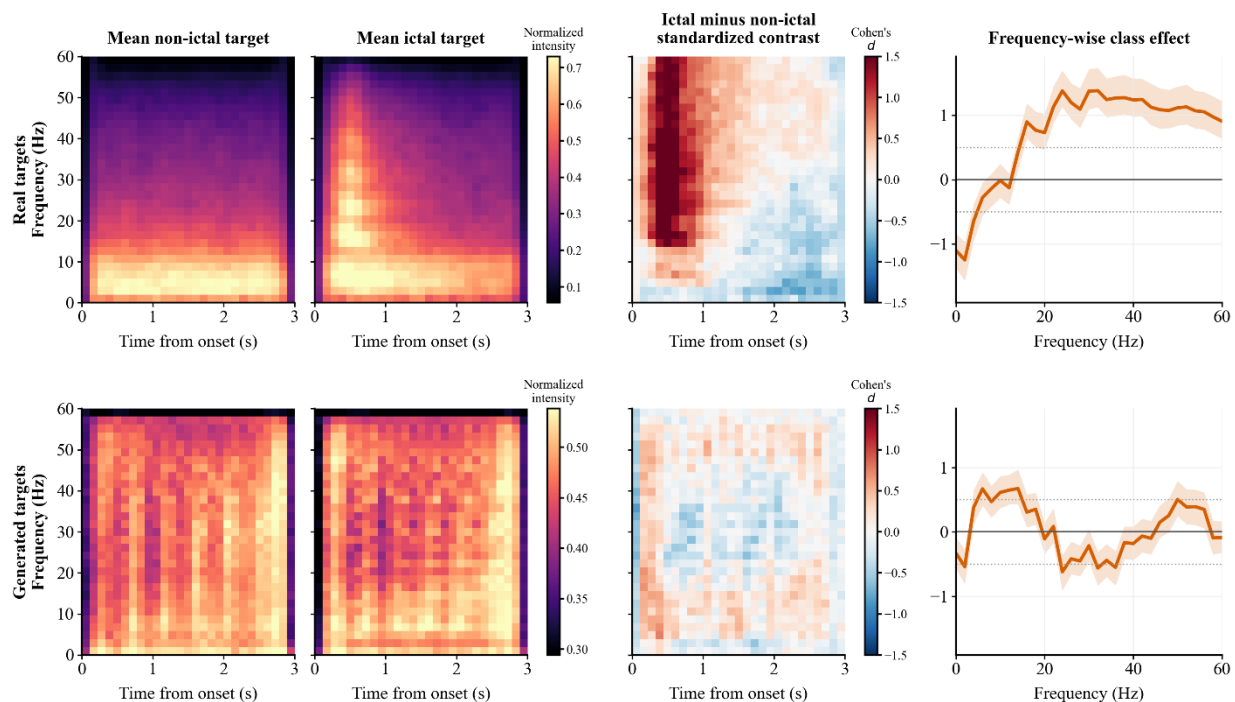


Figure 7. Class-conditional TC target structure at h0. The layout and aggregation protocol match Figure 6. Real TC targets show a strong early broadband ictal contrast. Generated targets retain a weaker frequency-dependent pattern, with positive effects at lower frequencies, negative effects over an intermediate band, and a smaller positive high-frequency effect.

Spectral contrasts do not show whether the generated targets preserve class separation in the representation used for scoring. We therefore compare real and generated targets in frozen-classifier feature space and examine their predicted ictal-probability distributions in Figure 8.

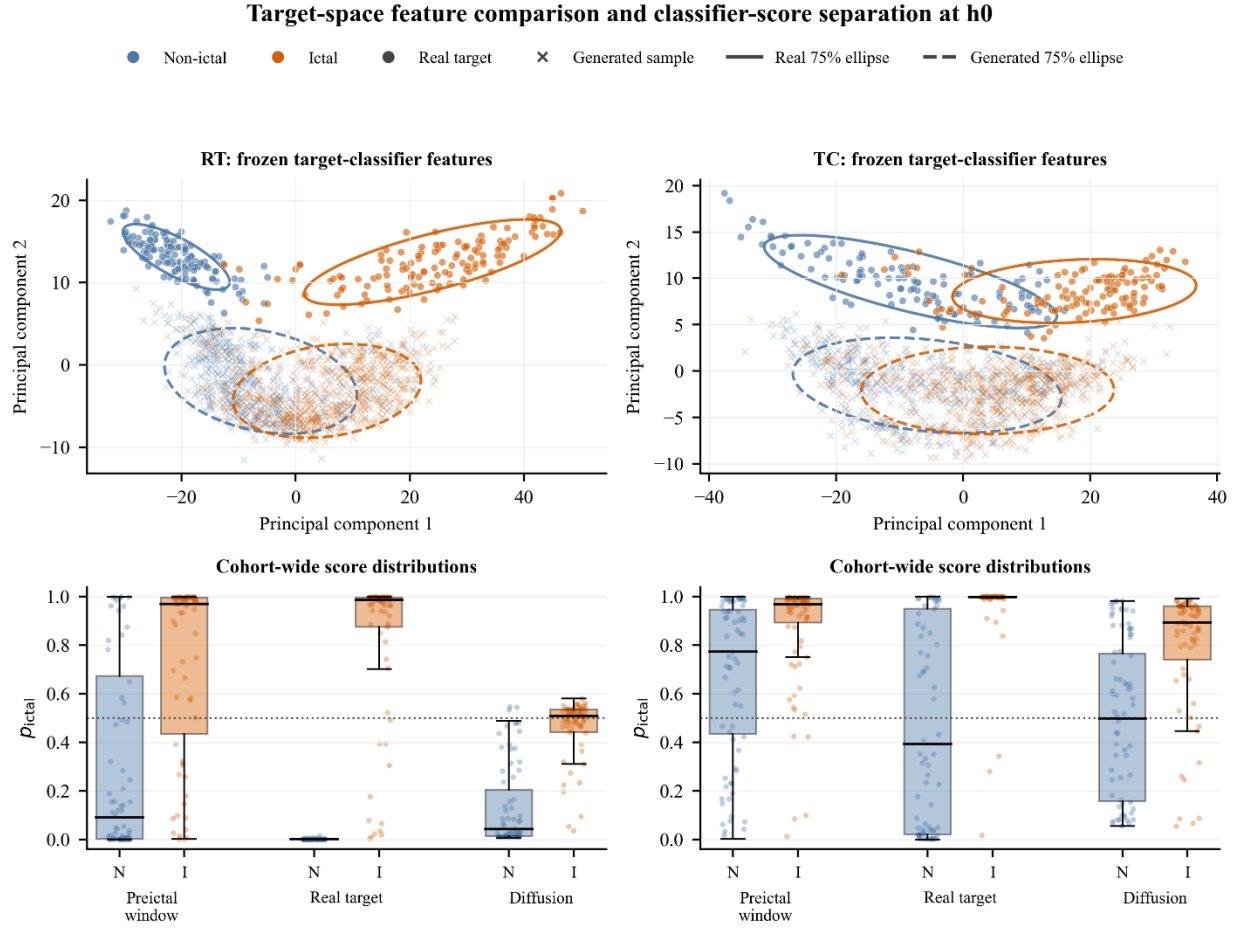


Figure 8. Target-space feature and score diagnostics at h0. Top: principal-component projections of frozen target-classifier features for real targets (circles, solid 75% ellipses) and one generated sample per event from each checkpoint (crosses, dashed 75% ellipses), colored by true class. Principal component analysis is fitted jointly within cohort after feature standardization. Bottom: test-set predicted ictal-probability distributions for the direct multimodal classifier, real-target classifier, and five-checkpoint diffusion ensemble. *N* and *I* denote non-ictal and ictal events, respectively.

The first two principal components explain 50.6% of standardized feature variance in RT and 44.7% in TC (Figure 8). Real and generated targets occupy visibly different regions, demonstrating that high forecasting AUC does not imply distributional or paired-reconstruction fidelity. Within the generated domain, however, classifier scores shift in the expected direction: non-ictal and ictal medians are 0.042 and 0.509 in RT and 0.498 and 0.894 in TC. Together with the standardized spectrogram contrasts, these results show that generated targets encode class-relevant time-frequency structure that is amplified by the nonlinear target classifier.

Generated contrasts remain attenuated relative to real targets, and individual samples should not be interpreted as reconstructions of the observed post-onset realization. The evidence therefore supports task-relevant target-space forecasting, not physiological or pixel-level generative fidelity. Future generative-quality evaluation should include held-out spectral distribution distances and classifier-independent neurophysiological features before generated targets are interpreted mechanistically.

4 Discussion

4.1 Interpretation

The paired comparisons separate the effect of target-space generation from the effect of adding modalities. Full multimodal diffusion consistently exceeds the matched full multimodal direct classifier in both cohorts, supporting the central hypothesis that a generated onset-adjacent representation can be easier to read out than the original preictal window. In TC, however, full multimodal diffusion is weaker than EEG-only diffusion at h1-h5. The evidence therefore supports target-space forecasting, but not the assumption that every available modality should always be fused.

The nonmonotonic horizon profiles also argue against a simple near-onset detector. Both cohorts retain predictive signal at earlier horizons, but the relative value of EEG, EMG, and photometry changes with horizon and cohort. The stronger RT performance is consistent with more stable preictal seizure-related structure in the RT recordings, whereas TC multimodal fusion may be more sensitive to modality noise or subject shift. These interpretations are hypotheses suggested by the current data and require confirmation in larger neurophysiological cohorts.

4.2 Relation to prior work

Most seizure-forecasting systems estimate risk directly from preictal recordings, with long-term monitoring, shared benchmarks, and deep EEG models progressively improving validation and representation learning (Mormann et al., 2007; Cook et al., 2013; Brinkmann et al., 2016; Kuhlmann et al., 2018; Truong et al., 2018). Our setting differs in two respects: evaluation is event-level and mouse-disjoint, and the primary model is generative rather than directly discriminative. The matched preictal classifier therefore remains the most informative baseline for isolating the effect of target-space generation.

Diffusion models have also been used for probabilistic time-series forecasting and imputation (Rasul et al., 2021; Tashiro et al., 2021), neurophysiological signal generation and augmentation (Vetter et al., 2024; Torma & Szegletes, 2025), and multi-channel EEG seizure warning (Jiang et al., 2026). Our method uses diffusion for a different purpose: it conditions generation on a preictal window and treats the generated onset-adjacent EEG representation as an intermediate forecast that is read out by a frozen classifier.

4.3 Limitations

The evidence is based on two fixed test mice and should be interpreted as a subject-disjoint proof of concept rather than a cohort-wide performance estimate. Replication across additional animals, recording sessions, and seizure models is needed to quantify between-subject variance and establish generalizability.

The TC results also show that multimodal fusion remains unresolved. EEG-only diffusion can outperform full multimodal diffusion, suggesting that EMG and photometry may introduce noise, distribution shift, or region-specific confounding. Future studies should test modality dropout, learned reliability gating, uncertainty-weighted fusion, and separate modality encoders.

The method also depends on a frozen target classifier. This design isolates target-space scoring, but a poorly calibrated or weak target classifier would limit the entire generated-target pipeline. A fuller evaluation should therefore include target-classifier calibration, component and loss ablations, guidance-scale sweeps, ensemble-size ablations, and external-subject replication.

Finally, the event database is balanced by construction and uses matched non-ictal windows sampled away from seizure onsets. This controlled setting is appropriate for method development, but deployment would require prevalence-aware calibration, false-alarm and time-in-warning analyses, and prospective temporal evaluation.

5 Conclusion

This study introduces target-space diffusion for preictal seizure forecasting. The method generates an onset-adjacent EEG representation from a preictal EEG or multimodal window and scores it with a frozen target classifier. In the fixed RT/M238 and TC/M710 evaluations, full multimodal diffusion improves over the matched full multimodal preictal classifier at every horizon, with mean gains of +0.063 and +0.089 AUC, respectively. EEG-only diffusion is also competitive and often surpasses full multimodal diffusion in TC/M710. The main insight is therefore not that more modalities always improve forecasting, but that target-space generation provides a promising representation whose robustness depends on modality reliability and mouse-disjoint generalization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Zanxu Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft, Visualization. Yunhao Luo: Methodology, Validation, Formal analysis, Investigation. Yueqing Peng: Conceptualization, Supervision, Writing - review and editing, Funding acquisition. Xiaofu He: Conceptualization, Supervision, Project administration, Writing - review and editing, Funding acquisition.

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Data and code availability

The raw data are available from the corresponding authors upon reasonable request.

The analysis code is available on GitHub [repository URL to be added before submission].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During manuscript preparation, the authors used OpenAI Codex to improve language, clarity, organization, and document formatting. The authors reviewed and edited all revised content and take full responsibility for the manuscript.

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